

# Proposition I.4

## The Side–Angle–Side Congruence Theorem

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### 1 Introduction

Euclid’s fourth proposition is the first theorem on triangle congruence.

Unlike the preceding propositions, it does not describe a geometric construction. Instead, it establishes that two triangles are congruent whenever two corresponding sides and the included angle are equal.

Euclid’s original proof relies on the method of superposition, one of the most discussed arguments in the *Elements*. Modern axiomatic systems generally avoid superposition as a primitive method of proof.

In the present reconstruction over Hilbert geometry, the proposition is obtained from the Side–Angle–Side (SAS) congruence theorem, which has already been established within the Hilbert theory.

Thus, while the mathematical content coincides with Euclid’s Proposition I.4, the logical structure of the proof is fundamentally different.

### 2 The Classical Configuration

Unlike the first three propositions of Book I, Proposition I.4 is not a geometric construction. Instead, it establishes a criterion for the congruence of two triangles.

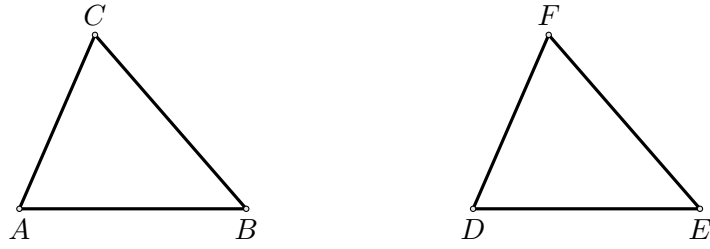
The proposition considers two triangles satisfying three geometric hypotheses:

- two pairs of corresponding sides are congruent;
- the included angles are congruent.

The objective is to prove that the triangles are congruent. As a consequence, the remaining corresponding side and the remaining corresponding angles are also congruent.

#### 2.1 Final Configuration

The geometric configuration is shown in Figure 1.



*Configuration of Proposition I.4.*

The hypotheses are

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

The desired conclusions are

$$BC \cong EF,$$

together with

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

### 3 Statement

#### **Proposition I.4.**

If two triangles have two corresponding sides congruent and the included angles congruent, then the triangles are congruent.

More precisely, let  $ABC$  and  $DEF$  be two non degenerate triangles:

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

## 4 Classical Proof

Euclid's original proof proceeds by superposition. One triangle is conceived as being placed upon the other in order to establish the coincidence of the corresponding sides and vertices.

The present reconstruction follows a different logical route. Superposition is not part of Hilbert's axiomatic system and therefore does not appear in the formal development.

Instead, the Side–Angle–Side congruence theorem has already been established within the Hilbert theory.

Applying the SAS theorem to the hypotheses

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF$$

immediately yields the congruence of the two triangles.

Consequently,

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This is precisely the conclusion of Proposition I.4.

□

## 5 Proof Architecture of the Reconstruction

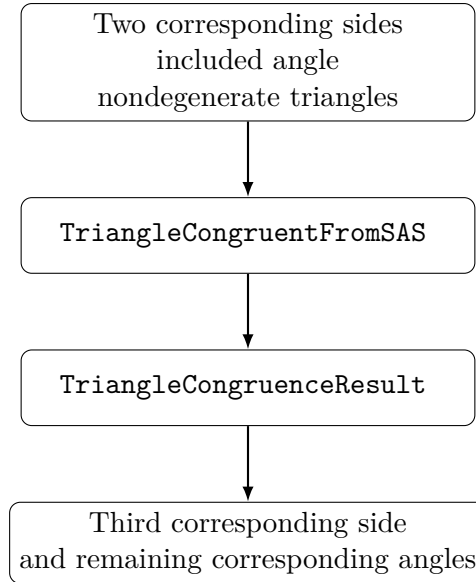


Figure 1: Logical structure of Proposition I.4 in the Hilbert reconstruction.

## 6 Hilbert Reconstruction

The principal difference between Euclid’s original argument and the present reconstruction concerns the method of proof rather than the mathematical statement.

Euclid establishes Proposition I.4 by superposition, treating one triangle as if it could be placed upon the other. Although historically important, this method is not available within the Hilbert axiomatic framework.

Instead, the present development relies on the Side–Angle–Side congruence theorem, which has already been established as part of the Hilbert theory.

Once the hypotheses of the SAS theorem have been established, an application of `TriangleCongruentFromSAS` immediately yields the complete triangle congruence.

Thus the classical proposition is reconstructed without appealing to superposition while preserving exactly the same mathematical content.

## 7 Lean Formalization

The Lean theorem is a direct application of the previously established Hilbert SAS theorem, exposed through the compatibility theorem `TriangleCongruentFromSAS`.

In addition to the two side congruences and the congruence of the included angles, the formal statement explicitly records that both triples of vertices are noncollinear.

The theorem returns a value of type `TriangleCongruenceResult`. This structure contains the congruence of all three corresponding sides and all three corresponding angles.

Thus the formal proof of Proposition I.4 consists of a single application of `TriangleCongruentFromSAS`; no additional geometric argument is required.

From the viewpoint of formalization, Proposition I.4 is also the first result whose Lean proof is essentially a single application of a previously established interface theorem.

## 8 Relationship with Earlier Propositions

### 8.1 Proposition I.2

The classical proof of SAS is expressed by Euclid through superposition rather than by a sequence of earlier construction propositions. Nevertheless, the preceding segment-transport theorem already establishes the general idea that congruent lengths can be reproduced independently of their original position.

In the Hilbert reconstruction, segment congruence is part of the foundational congruence structure, so no separate segment-copying construction is needed inside I.4.

### 8.2 Proposition I.3

Proposition I.3 develops the controlled cutting of one segment by a length congruent to another. It belongs to the same early congruence infrastructure, although it is not the decisive logical ingredient in SAS.

The contrast is useful: I.2 and I.3 are construction theorems about segments, whereas I.4 is the first major rigidity theorem for complete triangles.

### 8.3 Transition to Proposition I.5

Proposition I.5 uses SAS repeatedly. The auxiliary construction in the isosceles-triangle proof creates pairs of triangles with two corresponding sides and the included angle congruent, and I.4 propagates that local symmetry into further side and angle congruences.

Thus I.4 becomes one of the principal reusable congruence principles of Book I immediately after it is established.

## 9 Hilbert and Book Zero Components Used

### 9.1 Hilbert SAS

The decisive formal ingredient is the Hilbert SAS congruence axiom or its established interface theorem. Given two corresponding side congruences and congruence of the included angles, it supplies the remaining corresponding side and angle congruences.

The public Euclidean theorem therefore packages a foundational Hilbert congruence principle into the form of Proposition I.4.

## 9.2 Nondegenerate Triangles

The formal hypotheses explicitly require genuine triangles. The noncollinearity assumptions prevent the angle data from degenerating and make precise a condition that the classical diagram leaves implicit.

## 9.3 Angle Orientation

Angles are represented by ordered triples of points. Consequently the formal proof must ensure that the included angles supplied to SAS and the corresponding angles returned from it have exactly the orientations required by the Euclidean statement.

Symmetry and transport lemmas handle these changes explicitly.

## 9.4 Segment Congruence

The side hypotheses are expressed directly by the Hilbert segment congruence relation. No numerical length function or coordinate representation is introduced.

# 10 Formalization Notes

Proposition I.4 occupies a special position in the reconstruction.

For Euclid, SAS is proved by superposition: one triangle is imagined as being placed on another so that a side and the included angle coincide. The remaining equal side then forces the third vertex and hence the whole triangle to coincide.

In a Hilbert-style axiomatization, the same rigidity is foundational. SAS belongs to the congruence axioms. The Lean proof of the Euclidean proposition is therefore much shorter than Euclid's argument because the geometric content has already been placed in the Hilbert interface.

This is not a loss of information. It records a genuine difference in foundational organization:

Euclid: superposition proof      versus      Hilbert: SAS congruence principle.

The formal statement also makes nondegeneracy explicit. A diagram of two triangles silently suggests that each triple of vertices is noncollinear; Lean requires this information as part of the hypotheses needed to apply the angle-congruence machinery.

Proposition I.4 is therefore an early example where the Book I order and the logical order of the Hilbert foundation diverge sharply. What is a theorem for Euclid is already foundational infrastructure for the reconstruction.

# 11 Dependency Summary

The Euclidean presentation has the conceptual form

$$AB \cong DE, \quad AC \cong DF, \quad \angle BAC \cong \angle EDF$$

↓

superpose one triangle on the other

↓

the third vertices coincide

↓

$$BC \cong EF$$

and corresponding angles are congruent

↓

I.4.

The Hilbert reconstruction is structurally shorter:

side congruences + included-angle congruence

↓

Hilbert SAS

↓

third-side and corresponding-angle congruences

↓

euclid\_proposition\_4.

No new geometric axiom is introduced locally: the proposition exposes the SAS principle already present in the Hilbert congruence layer.

## 12 Conclusion

Proposition I.4 is the first fundamental triangle-congruence theorem of Book I.

Its historical proof uses superposition, one of the most discussed features of Euclid’s early geometry. In the Hilbert reconstruction the same mathematical rigidity is represented directly by the SAS congruence principle.

The resulting Lean theorem is therefore short for a substantive reason: the proof burden has moved from Book I into the axiomatic foundation.

This distinction becomes increasingly important in the propositions that follow. I.5 uses SAS as working infrastructure, and later constructions repeatedly rely on the ability to convert local side and angle congruences into congruence information about an entire triangle.

Proposition I.4 thus marks the point where the early segment constructions of I.1–I.3 give way to the systematic theory of triangle rigidity.